PROJECT REPORT OF INDUSTRY ORIENTED HANDS ON EXPERIENCE (IOHE)

ON

Reducing Hallucinations in AI using Retrieval Augmented Generation (RAG)

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

We hereby certify that the work which is being presented in the project report entitled **"Reducing Hallucinations in AI using RAG"** in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of our own work carried out under the supervision of **Dr. Gifty Gupta**. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

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This is to certify that the above statement made by the candidate is correct to the best of our knowledge and belief.

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ACKNOWLEDGEMENT

It is our pleasure to be indebted to various people, who directly or indirectly contributed in the development of this work and who influenced our thinking, behaviour and acts during the course of study. We express our sincere gratitude to all for providing me an opportunity to undergo Integrated Project as the part of the curriculum. We are thankful to Dr. Gifty Gupta for her support, cooperation, and motivation provided to us during the training for constant inspiration, presence and blessings. We also extend our sincere appreciation to Dr. Gifty Gupta who provided her valuable suggestions and precious time in accomplishing our project report. Lastly, we would like to thank the almighty and our parents for their moral support and friends with whom we shared our day-to-day experience and received lots of suggestions that improve our quality of work.

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1. Abstract

Large Language Models (LLMs) are widely used for generating human-like text and answering complex queries across various applications. However, they often produce hallucinated responses, which are factually incorrect yet presented with high confidence, thereby limiting their reliability in real-world scenarios.

In this project, we propose a Retrieval-Augmented Generation (RAG) based system to address this issue by grounding responses in external knowledge sources. The proposed approach integrates a structured pipeline that includes domain-specific dataset creation, data preprocessing, semantic chunking, and advanced retrieval techniques. We utilize keyword-based search, vector similarity search, and a hybrid retrieval approach to ensure the selection of relevant and contextually meaningful information.

The retrieved content is then provided to a pre-trained language model as contextual input, enabling it to generate responses that are more accurate and aligned with factual data. To further evaluate the reliability of the system, we introduce a novel metric called Truthfulness Score, which measures the degree of alignment between the generated response and the retrieved context.

Experimental results demonstrate that the proposed system significantly reduces hallucinated outputs, improves factual accuracy, and enhances overall response reliability when compared to baseline language models. This work highlights the effectiveness of combining retrieval mechanisms with language generation for building more trustworthy AI systems.

2. Introduction

2.1 Overview

Large Language Models (LLMs) have significantly transformed the field of Natural Language Processing by enabling machines to generate human-like and context-aware text.

They are widely used in applications such as chatbots, virtual assistants, and intelligent learning systems. However, despite their capabilities, LLMs often produce incorrect or fabricated responses, known as hallucinations.

2.2 Problem Background

Hallucination occurs because LLMs rely on probabilistic patterns learned during training rather than accessing verified or real-time data. This leads to responses that may appear correct but lack factual accuracy.

2.3 Need of the Study

In critical domains such as academic systems, healthcare, and institutional decision-making, incorrect information can lead to serious consequences. Therefore, reducing hallucination is essential for improving the reliability and trustworthiness of AI systems.

2.4 Proposed Approach

Retrieval-Augmented Generation (RAG) addresses this issue by combining information retrieval with text generation. It allows the system to access external data and generate responses based on real and relevant information.

3. Methodology

3.1 Dataset Preparation

A domain-specific dataset is created using syllabus documents, institutional policies, and course materials. These documents serve as the knowledge base for the system.

3.2 Data Preprocessing and Chunking

The collected documents are pre-processed and divided into smaller segments called chunks.

- Fixed-length chunking divides text into equal parts
- Semantic chunking divides text based on meaning

Semantic chunking is more effective as it preserves contextual integrity.

3.3 Retrieval Techniques

The system uses multiple retrieval methods:

- Keyword-based retrieval
- Vector similarity search
- Hybrid retrieval

Hybrid retrieval provides better performance by combining precision and semantic understanding.

3.4 Response Generation

The retrieved content is passed to a pre-trained language model, which generates responses based on the provided context.

4. Tools and Technologies

4.1 Programming Language

- **Python**

  Used for implementing the complete system, including data preprocessing, retrieval mechanisms, and integration with machine learning models

4.2 Frameworks

- **Hugging Face Transformers**

  Utilized for accessing pre-trained language models and generating context-aware responses based on retrieved information.

4.3 Vector Databases

- **FAISS / Pinecone**

  Used to store and efficiently retrieve document embeddings for fast similarity-based search during the retrieval phase.

4.4 Libraries

- **NumPy, Pandas, Scikit-Learn**

  Used for data manipulation, preprocessing, and basic machine learning operations such as similarity calculations and evaluation.

4.5 Concepts Used

- **NLP, Embeddings, RAG Architecture**

  Natural Language Processing techniques are used for text understanding, embeddings are used to represent text semantically, and RAG architecture combines retrieval with generation to produce accurate responses.

5. Implementation

The system is implemented using a pipeline-based architecture to ensure structured, efficient, and scalable processing of user queries.

Initially, the user query is taken as input and undergoes basic preprocessing such as text normalization and cleaning to improve retrieval accuracy. The processed query is then converted into a suitable format (such as embeddings) for efficient comparison with stored data.

The system retrieves relevant documents from the knowledge base using a combination of keyword-based search and vector similarity techniques. These documents are then ranked based on their relevance scores, and the top-k most relevant results are selected.

The selected documents are combined to form a contextual input, which is passed to a pre-trained language model. This step ensures that the model has access to relevant and domain-specific information before generating a response.

The language model processes this context and generates a response that is coherent, context-aware, and aligned with the retrieved information. This significantly reduces the chances of hallucination.

Additionally, the system ensures efficient indexing and retrieval of document embeddings to maintain performance, even with larger datasets.

Overall, the implementation focuses on optimizing retrieval quality, improving context selection, and generating responses that are both accurate and grounded in external knowledge sources.

6. Major Findings

The experimental results demonstrate a significant reduction in hallucinated responses when using the proposed RAG-based system compared to baseline language models.

The system achieves improved factual accuracy, as the generated responses are grounded in retrieved and relevant contextual information rather than relying solely on internal model knowledge.

Hybrid retrieval outperforms individual retrieval methods by effectively combining the precision of keyword-based search with the semantic understanding of vector similarity search, resulting in more relevant document selection.
Semantic chunking further enhances contextual understanding by preserving meaningful information within text segments, which improves both retrieval quality and response generation.

A new evaluation metric, called Truthfulness Score, is introduced to measure the alignment between generated responses and the retrieved context.

This metric helps identify hallucinated or unsupported information and provides a quantitative measure of response reliability. Higher Truthfulness Scores indicate better grounding in source data and more trustworthy outputs.

7. Future Scope

The proposed system can be further enhanced in several ways:

- Integration of real-time data sources to ensure up-to-date and dynamic responses
- Implementation of advanced retrieval techniques for improved relevance and efficiency
- Expansion to multi-domain datasets for broader applicability
- Incorporation of automated fact-checking mechanisms to further reduce hallucinations
- Optimization for large-scale deployment and improved system performance

8. Conclusion

In this project, we demonstrated that Retrieval-Augmented Generation (RAG) is an effective approach for reducing hallucinations in large language models.

By integrating retrieval techniques, semantic chunking, and contextual response generation, we improved the factual accuracy and reliability of AI-generated outputs. The use of external knowledge grounding ensures that responses are aligned with relevant and verified information.

Additionally, the introduction of the Truthfulness Score provides a practical way to evaluate the alignment between generated responses and retrieved context, helping in identifying hallucinated content.

Overall, the proposed system contributes towards building more trustworthy, accurate, and context-aware AI systems suitable for real-world applications.

9. References

- Lewis et al., “Retrieval-Augmented Generation,” 2020
- Vaswani et al., “Attention is All You Need,” 2017
- FAISS Documentation
- Hugging Face Transformers Documentation